

Evaluating and Improving Cross-Dataset Generalization of Machine-Learning-Based Intrusion Detection Systems

B.Tech. Project – I | Mid-Semester Evaluation

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Introduction

- **Network Intrusion Detection Systems (NIDS)** monitor traffic to detect attacks.
- ML-based IDS learn attack patterns from labelled traffic and dominate current research.
- Standard benchmarks: **CIC-IDS-2017**, **UNSW-NB15**, **TON_IoT**.
- **Usual evaluation:** train and test on the *same* dataset → accuracy > 99%.
- **Reality:** an IDS is trained on one network and deployed on a *different* one.
- **Cross-dataset generalisation:** how well does a model trained on dataset A work on dataset B?

Motivation

- **Deployment reality:** organisations rarely have large labelled data for their own network — they must reuse models trained elsewhere.
- **The gap is severe:** independent studies show cross-dataset accuracy falling to <40% or near **random chance**.
- **It is a feature problem:** models rely on dataset-specific features (IPs, ports, collection-window statistics) instead of real attack behaviour.
- **Metaheuristic feature selection is mature** — but *always* optimises **same-dataset** accuracy.
- **Our idea:** re-aim the optimiser at **cross-dataset robustness** — simple, powerful, unexplored.

Problem Statement & Objectives

Problem statement

- ML-based IDS are evaluated with same-dataset splits, overstating real-world value.
- On a structurally different network, performance collapses [1], [2].
- Existing feature-selection methods optimise only same-dataset accuracy [8]–[12].
- **Need:** characterise the cross-dataset gap, and select features *for transfer*.

Objectives

1. Evaluate cross-dataset generalisation of ML/DL models across the three datasets.
2. Characterise which features and attack classes transfer vs. which are artefacts.
3. Compare metaheuristic optimisers (GA, PSO, ...) for transfer-oriented feature selection.
4. Benchmark against standard feature-selection baselines.
5. Produce practical guidelines for generalisable IDS.

The Three Datasets

Dataset	Source / tool	Records	Features	Classes
CIC-IDS-2017	CIC, CICFlowMeter	2,830,743	78	15
UNSW-NB15	ACCS, Argus + Bro	257,673	42	9 + Normal
TON_IoT	UNSW, Zeek (Bro)	211,043	42	10

- Collected on **different networks** with **different feature-extraction tools**.
- Different label vocabularies and very different class balances.
- This diversity is exactly what makes them suitable for a generalisation study.

Core Challenge: Feature-Schema Mismatch

- The three datasets share **no common column name**.
- CICFlowMeter statistics $\neq$ Argus/Bro counters $\neq$ raw Zeek log fields.
- Only **~6 raw features** are semantically shared (duration, bytes, packets...); ~15–25 after encoding + derived ratios.

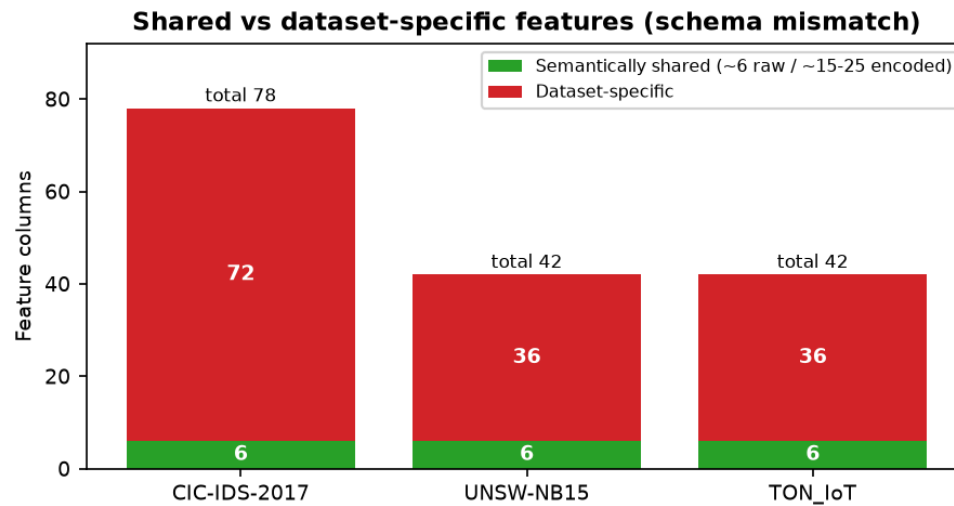


Figure 1: Shared vs dataset-specific features

This mismatch is the root cause of the generalisation gap — and the focus of our work.

Proposed Methodology (overview)

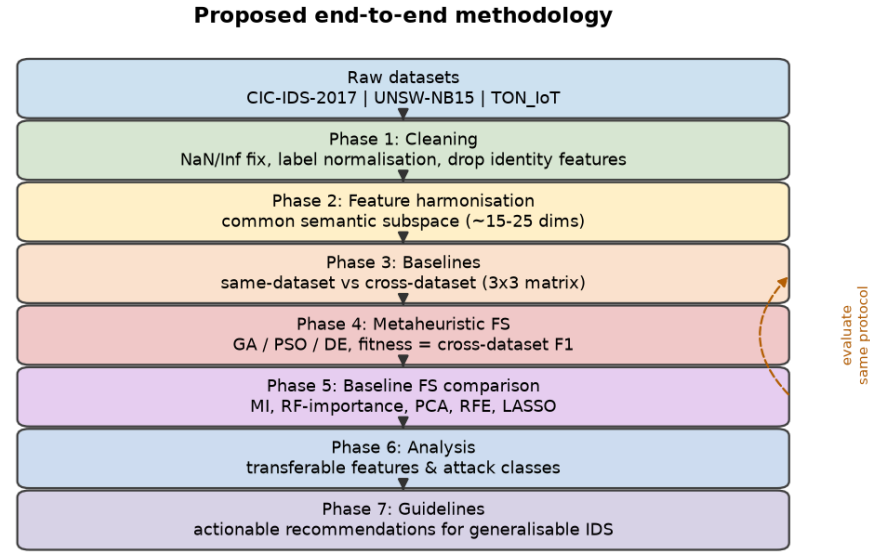


Figure 2: Proposed end-to-end methodology

Feature Harmonisation (shared subspace)

We map semantically equivalent features across datasets into one common subspace:

Concept	CIC-IDS-2017	UNSW-NB15	TON_IoT
Duration	Flow Duration	dur	duration
Source bytes	Total Length of Fwd Packets	sbytes	src_bytes
Dest bytes	Total Length of Bwd Packets	dbytes	dst_bytes
Source packets	Total Fwd Packets	spkts	src_pkts
Dest packets	Total Backward Packets	dpkts	dst_pkts
Protocol / Service / State	(encoded)	proto,service,state	proto,service,conn_state

- Plus derived ratios: bytes/packet, packets/second, forward/backward ratio.
- **Remove identity/leakage features** (IPs, ports, IDs).

Metaheuristic Feature Selection (core contribution)

- **Solution** = **binary mask** over the shared subspace (which features to keep).
- **Fitness** = **cross-dataset generalisation** (the novelty):

fitness(mask) = average balanced-accuracy / macro-F1
 of model trained on source[mask]
 and evaluated on target[mask]
 over all source→target pairs

- **Optimisers compared:** Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution / Whale Optimization.
- Optional multi-objective NSGA-II: maximise cross-dataset F1 **and** minimise number of features.
- **Key difference from prior work:** the optimiser is rewarded for *transfer*, not for fitting one dataset.

Evaluation Protocol & Baselines

Protocol

- *Same-dataset*: stratified 5-fold cross-validation.
- *Cross-dataset*: train on source, test on target $\rightarrow$ full 3×3 matrix.
- *Metrics*: balanced accuracy, macro-F1, per-class F1 (not overall accuracy — imbalance is dataset-specific).
- *Models*: RF, XGBoost/LightGBM, SVM, Logistic Regression, small MLP.
- Fixed seeds, repeated runs for reproducibility.

Baselines for feature selection

- No feature selection; mutual information; random-forest importance; PCA; RFE; LASSO — all matched to the same subset size as the metaheuristic.

Literature Review (1/2)

#	Title	Author(s)	What it does	Conclusion
1	Cross-Dataset Generalization of ML for NIDS	Cantone et al., 2024	Cross-dataset benchmark (4 sets)	Near-chance cross-dataset; diagnosis only
2	Assessing Generalisation Capability	Hossain et al., 2026	RF/LR/NB on UNSW ↔ TON	<40% cross-dataset vs 95–99% in-dataset
3	Cross-Domain Failure in Lightweight IDS	Hakim et al., 2026	3 IIoT sets + SHAP	Port shortcut; protocol reverses results
4	Tabular Representation Learning for NIDS	Butt et al., 2026	Representation learning	Transfer depends on source→target pair
5	CIC-IDS-2017	Sharafaldin et al., 2018	Dataset	Primary dataset
6	UNSW-NB15	Moustafa & Slay, 2015	Dataset	Second dataset
7	TON_IoT	Moustafa, 2021	Dataset	Third dataset (Zeek)
8	Metaheuristic classifiers for ID	Fatima & Ali, 2022	GA FS + ensembles	Strong single-dataset; no transfer

Literature Review (2/2)

#	Title	Author(s)	What it does	Conclusion
9	Unsupervised FS via NSGA-II	Suman et al., 2019	Multi-objective FS	Pareto subsets; single-dataset
10	MPDGGA	Li, 2026	Multi-population GA FS	Best 10/11 datasets; single-dataset
11	GA + Whale Optimization FS	Mojtahedi et al., 2022	Hybrid bio-inspired FS	Swarm hybridisation viable
12	NSGA-III AutoML IDS	Yang et al., 2026	Multi-objective AutoML	Efficiency, not transfer
13	DI-NIDS	Layeghy et al., 2022	Domain-invariant features	Better cross-domain; needs target data
14	Survey of NIDS Data Sets	Ring et al., 2019	Dataset taxonomy	Explains heterogeneity
15	Adversarial Robustness (NewCICIDS)	Vitorino et al., 2024	Corrected CICIDS2017	Corrected data changes conclusions

Research Gap

Group	Cross-dataset eval?	Metaheuristic FS?	Optimised for transfer ?
Diagnostic studies [1]–[4]	Yes	No	No
Metaheuristic FS [8]–[12]	No	Yes	No (same-dataset fitness)
Domain adaptation [13]	Yes	No	No (needs target data)
This project	Yes (3 datasets)	Yes (GA, PSO, ...)	Yes

- **No published work selects features using a cross-dataset generalisation objective for NIDS.**
- Our project fills exactly this gap.

Implementation & Progress

Completed

- Reproducible Python environment + version-controlled repository.
- All three datasets downloaded and analysed; figures generated automatically.
- **Shared 12-feature subspace implemented** (identity/leakage features removed).
- **Cross-dataset baseline benchmark: 5 ensemble models trained on the full datasets**, evaluated on all pairs $\rightarrow 3 \times 3$ matrices.

Next (end-semester)

- GA / PSO / DE optimisers with the cross-dataset fitness function.
- Standard feature-selection baselines (MI, RF-importance, PCA, RFE, LASSO).
- Transfer analysis (SHAP/UMAP, ablations) and practical guidelines.

Baseline Cross-Dataset Results

- Trained **5 ensemble models on the full datasets**; tested on all three (3×3 matrix).
- Same-dataset accuracy **0.96–0.97**; cross-dataset only **0.37–0.42**.
- Cross-dataset **balanced accuracy 0.46–0.49** → at/near random chance (0.50).

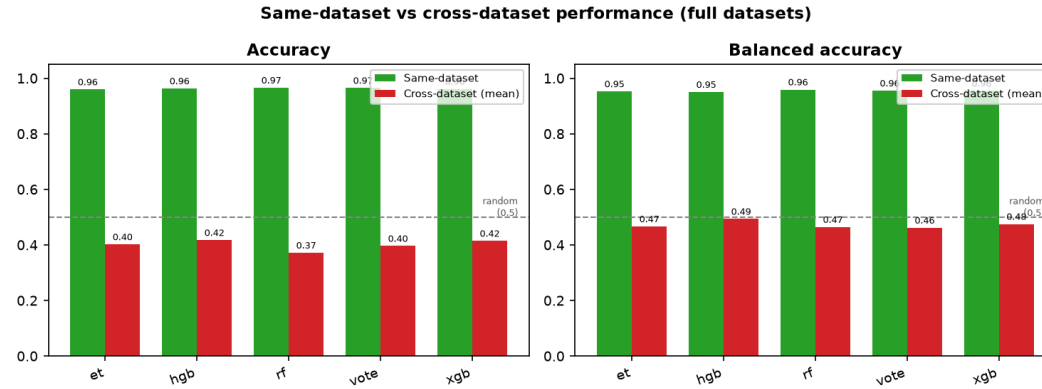


Figure 3: Same-dataset vs cross-dataset

Baseline Results — Key Findings

Model	Same-dataset Acc	Cross-dataset Acc	Cross-dataset Bal-Acc
Random Forest (rf)	0.9666	0.3726	0.4650
XGBoost (xgb)	0.9622	0.4150	0.4760
Hist. Grad. Boosting (hgb)	0.9638	0.4179	0.4941
Extra Trees (et)	0.9602	0.4027	0.4672
Soft-voting (vote)	0.9656	0.3968	0.4624

- XGBoost CIC-IDS-2017 → UNSW-NB15 balanced accuracy = **0.5000 (exactly random)**.
- **Ensembling does NOT fix the gap** → the problem is the features/distribution, not model capacity.
- Independently matches the literature [1], [2]; motivates transfer-oriented optimisation.

DA-MFS — Domain-Aligned Metaheuristic Feature Selection

- **Why plain selection fails:** the same feature has a *different distribution* in each dataset.
- **Stage 1 — Domain alignment:** align each dataset’s feature marginals (per-dataset quantile transform $\rightarrow$ normal).
- **Stage 2 — Metaheuristic:** GA / binary PSO search a feature subset with a **cross-dataset fitness** (mean balanced-accuracy + macro-F1 over 6 pairs).
- **Novelty:** no prior work couples domain alignment with metaheuristic feature selection for NIDS.

Cross-dataset metric	Raw baseline	DA-MFS	Change
Accuracy	0.373	0.508	+0.136
Balanced accuracy	0.465	0.569	+0.104
Macro-F1	0.308	0.460	+0.153
Attack recall	0.479	0.607	+0.128

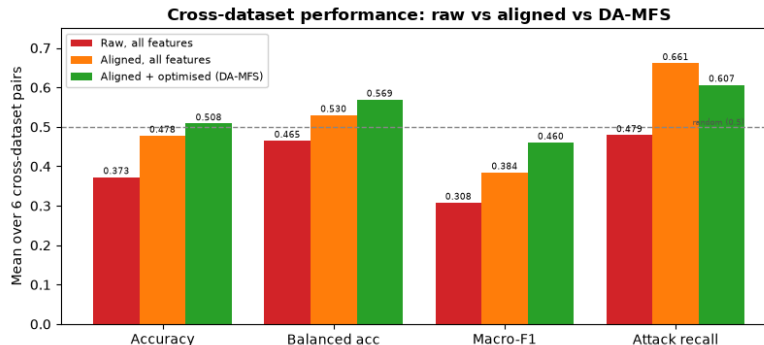


Figure 4: Raw vs aligned vs DA-MFS

- Compact **4-feature** subset; **+13.6 pp accuracy and +12.8 pp recall** — a clear, defensible gain.
- Next: GA vs PSO vs DE, NSGA-II multi-objective, divergence-aware fitness, standard FS baselines.

Gantt Chart

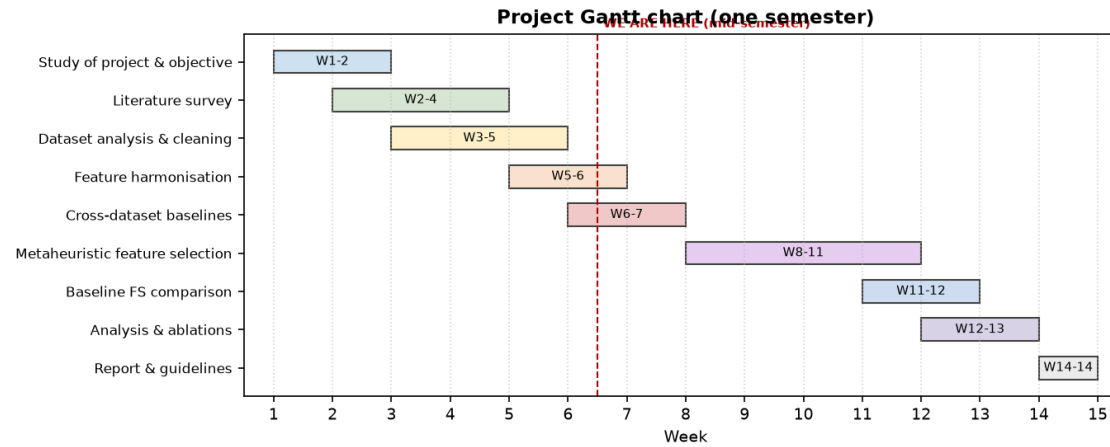


Figure 5: Project Gantt chart

Conclusion & Future Work

Conclusion

- Three structurally different benchmarks analysed; **feature-schema mismatch** is the root cause of cross-dataset failure.
- Literature review (15 core papers) confirms a clear gap: no transfer-optimised metaheuristic feature selection for NIDS.
- **Baseline quantified the gap:** 0.96–0.97 same-dataset vs 0.37–0.42 cross-dataset (balanced accuracy $\approx$ random chance).
- **DA-MFS optimisation works:** domain alignment + evolutionary feature selection lifts cross-dataset accuracy **0.373 $\rightarrow$ 0.508** and recall **0.479 $\rightarrow$ 0.607**.

Future work

1. Extend DA-MFS (GA / PSO / DE / NSGA-II, divergence-aware fitness).
2. Comparison vs. standard feature-selection baselines.
3. Analysis of transferable features/attacks + ablations.
4. Practical guidelines and final report.

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